SKY130 Area Estimator: Model Comparison

1. The Old Model (Flawed)

The previous script fitted a simple linear regression to calculate the layout area of MOSFETs:

Area = a*(W * nf) + b*L + c

This model is mathematically incomplete. In a physical Magic layout, the overall area is a rectangle defined by its bounding box (Width × Height). Height is proportional to W (due to source/drain extensions), while Width is proportional to (L * nf) (due to multiple gates). Therefore, the true Area equation MUST have a multiplication cross-term (W × L), which the simple linear model lacked.

2. The New Model (Accurate Bounding-Box)

To solve this, the new script separately models the bounding box dimensions based on the device_db data:

Height = ah*W + bh
Width = aw*(L * nf) + bw*nf + cw

Then, the final Area is calculated properly as:

Area = Height × Width

This correctly captures the cross-multiplication term, allowing accurate scaling for devices that are simultaneously wide and long.

3. Numerical Error Comparison (nfet_01v8)

Dimensions	Actual Area (μm^2)	Old Model Prediction	New Model Prediction
W=0.42 L=0.15 nf=1	5.32	6.06 (14.0% err)	5.34 (0.5% err)
W=1.00 L=0.15 nf=1	6.54	6.68 (2.1% err)	6.57 (0.5% err)
W=2.00 L=0.15 nf=4	14.72	14.11 (4.1% err)	14.72 (0.0% err)
W=2.00 L=0.35 nf=2	12.10	10.78 (10.8% err)	12.19 (0.8% err)
W=0.42 L=0.50 nf=1	6.20	7.67 (23.8% err)	6.15 (0.9% err)
W=1.00 L=0.50 nf=2	10.07	9.35 (7.2% err)	10.07 (0.1% err)

As shown above, the new model reduces worst-case estimation errors from nearly 24% down to under 1%.

4. Parity Plots

The plots below show Actual Area (X-axis) vs Predicted Area (Y-axis). A perfect model lies exactly on the dashed black line. The new model (blue circles) follows the ideal line perfectly, whereas the old model (red crosses) diverges heavily.